



Web Technologies
Semester Project Report

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Submitted To:

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Game Vault

Web Scraper App_ Full Stack Game Discovery and Management

Abstract

This project is a full-stack web application developed using Angular, Node.js, Express.js, and MongoDB. The system allows users to discover, search, manage, and save game information obtained from external APIs and web scraping services. It provides secure user authentication, responsive design, and personalized game collections.

Objectives

Develop a modern web application using current web technologies; implement secure user authentication using JWT; fetch and display game data from external sources; allow users to save and manage game collections; provide a responsive and user-friendly interface.

Technologies Used

Frontend: Angular 16, TypeScript, Bootstrap 5, RxJS. Backend: Node.js, Express.js. Database: MongoDB with Mongoose. Security: JWT Authentication, Helmet, Express Rate Limiter. Other Tools: Axios, Cheerio, CORS, Morgan.

System Architecture

The application follows a three-tier architecture. The Angular frontend communicates with the Express backend through REST APIs. The backend handles authentication, business logic, web scraping, and database operations. MongoDB stores user and game data.

Modules

1. User Registration and Login.
2. JWT-based Authentication.
3. Game Search and Filtering.
4. Game Data Scraping/API Integration.
5. Saved Games Management.
6. User Profile Management.
7. REST API Services.

Database

MongoDB is used as the primary database. Mongoose ODM provides schema management and data validation. User information, authentication details, and saved game collections are stored in the database.

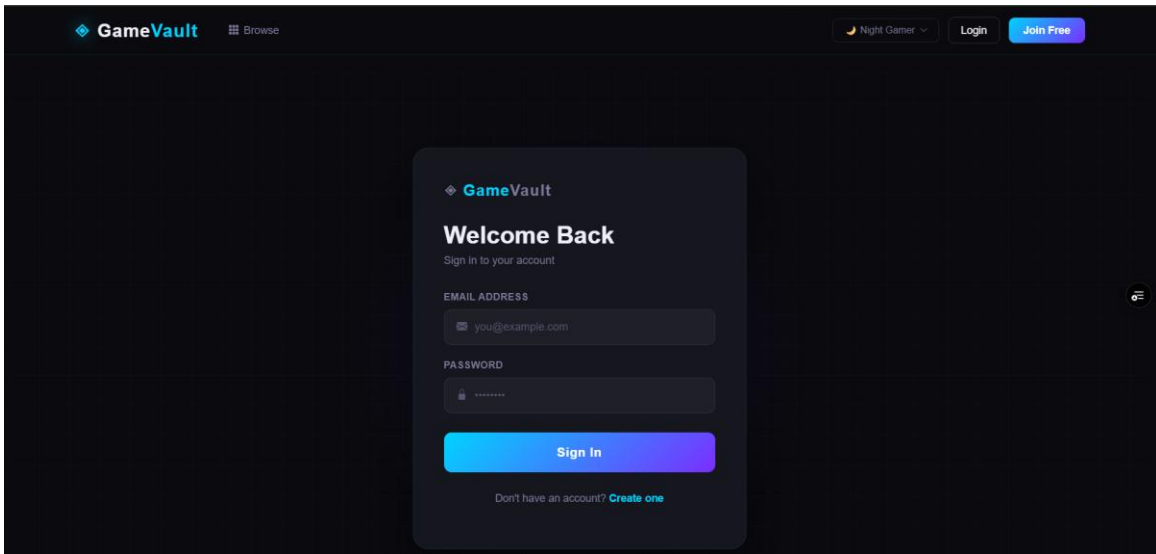
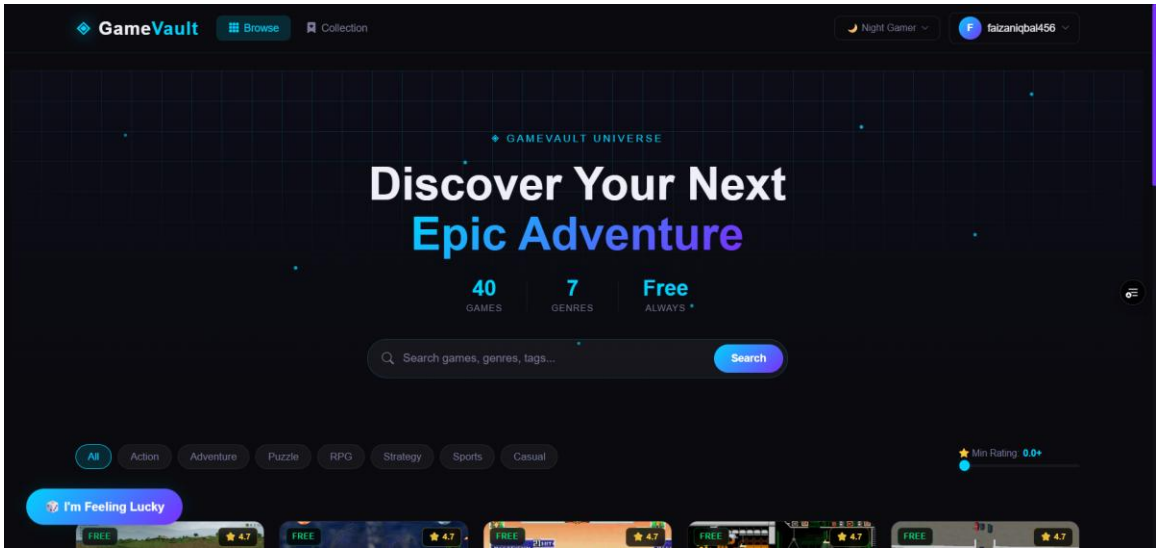
Security Features

Password hashing using bcryptjs, JWT token authentication, rate limiting against abuse, Helmet security headers, input validation, and controlled API access.

Working Procedure

Users create an account and log in. The frontend sends requests to backend APIs. The backend retrieves game information through scraping/API services and stores relevant data in MongoDB. Users can browse, search, save, and manage games through the interface.

(Detailed videos of working are on github)



GameVault

Create Account

Join thousands of gamers

USERNAME

coolplayer99

EMAIL ADDRESS

you@example.com

PASSWORD

CONFIRM PASSWORD

FIRST NAME (optional)

John

LAST NAME (optional)

Doe

Create Account

Already have an account? [Sign in](#)

GameVault

Browse

Collection

My Game Collection

All (1)

Wishlist (1)

Playing (1)

Completed (1)

Archived (1)



Jump Ultimate Stars

Wishlist

Fighting

View Details

Remove

Night Gamer

Night Gamer

Light Mode

Neon Mode

Chill Blue

Cyberpunk

Advantages

Responsive design, scalable architecture, secure authentication, easy maintenance, separation of frontend and backend, and modern technology stack.

Contribution:

Member	Responsibilities
Faizan Iqbal	Project planning, requirements gathering, website structure design, coordination of team activities, report compilation and final documentation.
Khaleeq Waqar	Front-end development including HTML, CSS, Bootstrap, user interface design, page layouts, and responsive design implementation.
Awais Sajjad	Back-end development, database integration, testing, debugging, deployment, and preparation of project presentation/demo.

Note

In order to check the code kindly enter the master branch instead of main branch (selected as the default) as all the codes are added in the master branch there.

(Video for this explanation is pushed on github)

Conclusion

The Web Scraper App successfully demonstrates the implementation of modern web technologies in a real-world application. It combines frontend development, backend services, database management, authentication, and web scraping into a single integrated solution.